



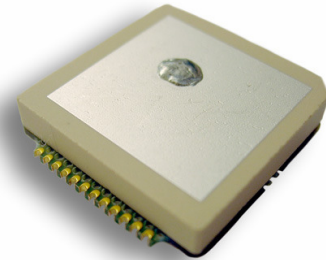
a time and place...
for everything

Wi-Sys Communications Inc.

WS5021 GPS Receiver + Antenna OEM Module

Applications:

- ◆ Portable GPS Tracking
- ◆ Automotive Navigation Systems
- ◆ Fleet Management & Telematics Systems
- ◆ Marine Navigation Systems
- ◆ Asset Tracking Equipment
- ◆ Personal Navigation
- ◆ Tracking Systems and Mapping Devices



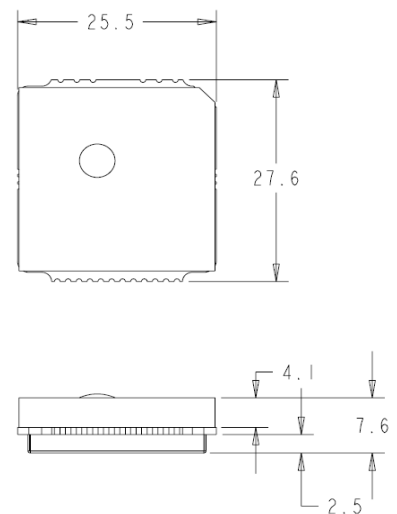
The WS5021 GPS Receiver + Antenna OEM Module incorporates a highly sensitive 12-channel, quick time-to-fix GPS Receiver and a high performance active GPS antenna for optimum signal reception. It features the newest generation super-integrated GPS System on Chip (SoC), a custom designed ceramic patch element that minimizes detuning effects caused by adjacent objects, as well as a Low Noise Amplifier and a SAW filter.

The WS5021 GPS Receiver + Antenna OEM Module is housed in a very compact 25.5 mm x 27.6 mm x 7.6 mm PCB surface mount package or a connectorized module and comes with 3.3V CMOS interface.

Features:

- ◆ Integrated GPS receiver, GPS antenna, LNA & SAW filter
- ◆ High performance GPS engine (-146 dBm sensitivity)
- ◆ 12 channel simultaneous operation
- ◆ Low noise LNA and SAW band-pass filter
- ◆ GPS output: NMEA 0183 output
- ◆ 1 PPS timing source
- ◆ 3D fix output
- ◆ Low current: 75mA
- ◆ 3.3V operation
- ◆ Flexible mounting options
- ◆ RoHS compliant
- ◆ Starter kit, and test software available

Dimensions: WS5021-SMT



Ordering Information:

WS5021-SMT: PCB SMT Package with 3.3Volt CMOS interface
WS5021-C: Connectorized package with 3.3Volt CMOS interface



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Specifications:

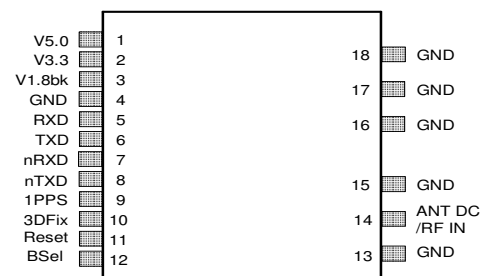
Mechanical	
Dimensions	25.5 mm x 27.6 mm x 7.6 mm
Interface	Reflow soldering, 3.3 V CMOS
Shock	Vertical axis 50G, other axis 30G
Vibration	3 axis sweep = 15 min (10-200 Hz log sweep)

Environmental	
Operating Temperature	-40 to +85 C
Storage	-45 to +85 C
Humidity	95% max (non-condensing)

Electrical	
Voltage	3.3V- 5.5V
Current	75 mA @ 3.3 V

Antenna Response	
Frequency	1575.42 MHz
Gain	@ 90° 3 dBic @ 20° -2 dBic
Polarization	Right Hand Circular
Axial Ratio	@ 90° 4 dB @ 20° 6 dB
Out-of-band rejection	+15 MHz: 10 dB/- 15 MHz: 30dB +20 MHz: 20 dB/- 20 MHz: 30dB +40 MHz: 40 dB/- 40 MHz: 40dB

GPS Performance	
Frequency	L1, 1575.42 MHz
Channels	12 channels parallel
Sensitivity (min)	-144 dBm (26dB-hz)
Position Accuracy	< 30 meters SPS < 1 metre DGPS
Time to First Fix	Autonomous start < 90 sec Cold start < 45 sec Warm start < 7sec Re-acquisition < 1 sec
Serial Protocol:	NMEA messages - 4800 Baud (default) GGA5, VTG, GSA, GSV, on UTC second - 57,600 baud: GGA5, GSA, GSV, RMC, TG, TS, PA, on UTC second



Specifications subject to change without notice
Please visit our website or contact us for additional information

Also Available:

GPS Engine Products

WS5011: Magnet-Mount GPS Receiver + Antenna with USB or CMOS interface

WS5121: 27.6 mm x 25.5 mm x 3.5 mm GPS Receiver OEM Module with CMOS interface

WS5071: 60 mm dia. Hard-Mount GPS Receiver + Antenna with USB or CMOS interface

GPS Timing Engine Products

WS5070: 60 mm dia. Hard-Mount GPS Timing Receiver + Antenna with RS233 or RS422 interface

WS5020: 27.6 mm x 25.5 mm x 7.6 mm GPS Timing Receiver + Antenna OEM Module with CMOS interface

WS5120: 27.6 mm x 25.5 mm x 3.5 mm GPS Timing Receiver OEM Module with CMOS interface

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